

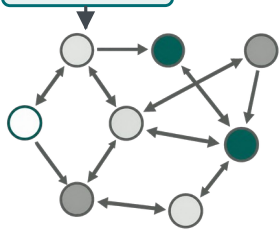
FLOW SPACE

budgets b_k

$b_k, k = 1 \dots K$

2 Flow graph

discretise



D

$\hat{F}_{\text{roll}}$

$\hat{\mu}_0$

$\omega^{(k)}, \bar{c}_k$

3 Capacity compile

$S^{(1)}$

$S^{(2)}$

$\vdots$

$S^{(K)}$

$S^{(1 \dots K)}$

$$S^{(k)} = b_k \omega^{(k)} / \bar{c}_k$$

4 Compose

hard min
(certificate)

softmin T_c

S
(capacity ceiling)

5 γ -Flow Sinkhorn

$$\min \langle D, F \rangle - \varepsilon H(F)$$

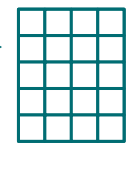
ε



row scaling



column scaling

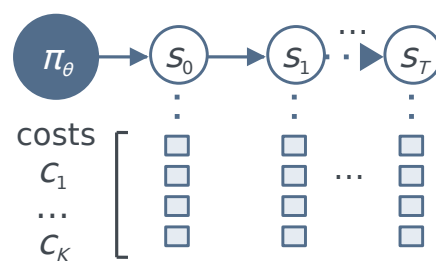


$$0 \leq F \leq S \quad F \mathbf{1} - \gamma F^T \mathbf{1} = (1 - \gamma) \mu_0$$

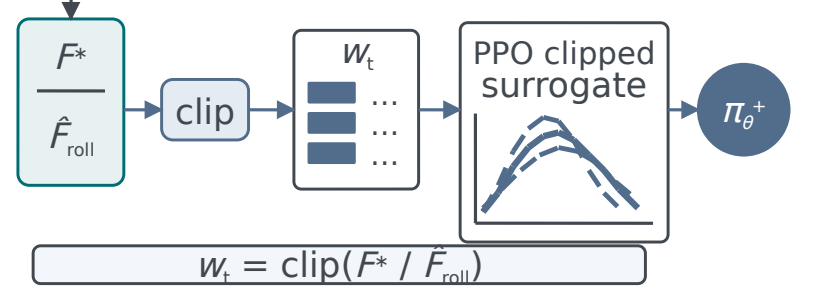
F^*

POLICY SPACE

1 Policy rollout



6 Flow-weighted PPO

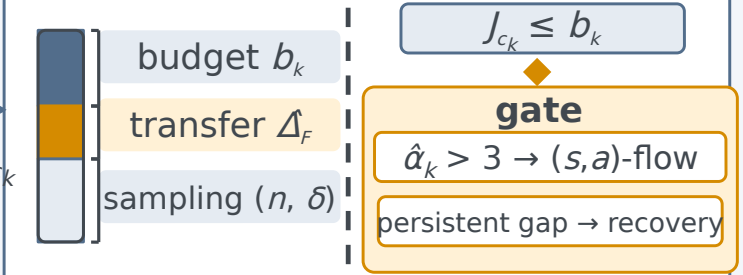


$$w_t = \text{clip}(F^* / \hat{F}_{\text{roll}})$$

π_{θ}^+

deployed policy

7 Deployed-policy audit



gate

$\hat{a}_k > 3 \rightarrow (s, a)\text{-flow}$

persistent gap $\rightarrow$ recovery

transitions, rewards,
 K costs

gates

F^*

next rollout